TITAN A16 - FEA SETUP GUIDE & QC PROTOCOL

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PART 1: FINITE ELEMENT ANALYSIS (FEA) SETUP

1.1 ANALYSIS OBJECTIVES

Primary Goal: Predict natural frequencies and mode shapes of TITAN A16 geometry

Target Validation:
- Identify resonance peak near **360 Hz ±10 Hz**
- Characterize mode shape at target frequency (torsional, longitudinal, or radial)
- Evaluate Q-factor (quality factor / damping ratio)
- Assess torsional deflection under 0.075mm constraint

1.2 SOFTWARE RECOMMENDATIONS

Preferred Platforms:
- ANSYS Mechanical (Modal analysis module)
- Abaqus/CAE (Frequency extraction)
- COMSOL Multiphysics (Structural Mechanics)
- SolidWorks Simulation Premium (Frequency study)

Minimum Requirements:
- Modal/frequency analysis capability
- Multi-material support
- Mesh control for curved features
- Result export (frequency response function)

1.3 GEOMETRY IMPORT

From CAD Model (STEP file):
1. Import TITAN A16 geometry preserving:
   - Central bore (Ø3.03mm)
   - Triple helix threads

  - Top lattice node pockets
  - Axial layer boundaries (if modeled as separate volumes)

2. **Simplification options** (if computation is too expensive):
  - Omit node pockets for initial modal study (add later for refinement)
  - Model helix as helical surface pattern rather than full thread geometry
  - Use axisymmetric model if appropriate (loses helix detail)

1.4 MATERIAL PROPERTY ASSIGNMENT

Method A: Homogenized Effective Properties (Simplified)

Use volume-weighted average properties assuming:
- 50% Tantalum
- 35% Titanium
- 12% Gold
- 3% Carbon

Effective Properties (Rule of Mixtures):

Property	Value	Units
Density (ρ)	11,200	kg/m^3
Young's Modulus (E)	145	GPa
Poisson's Ratio (ν)	0.34	—
Damping Ratio (ζ)	0.002 - 0.005	—

Method B: Multi-Material Zones (Accurate)

Assign individual material properties to axial layers:

Layer L1, L3, L5 (Titanium-dominant):
- Density: 4,510 kg/m^3
- Young's Modulus: 116 GPa
- Poisson's Ratio: 0.32

Layer L2, L4 (Gold-interfacial):
- Density: 19,320 kg/m^3
- Young's Modulus: 79 GPa
- Poisson's Ratio: 0.42

Core (if using discrete Ta anchor):
- Density: 16,650 kg/m^3

- Young's Modulus: 186 GPa
- Poisson's Ratio: 0.34

Interfaces:
- Use "bonded contact" or "tie constraint" between layers
- No slip assumed (perfect adhesion)

1.5 MESH GENERATION

Mesh Parameters:
Region	Element Type	Element Size	Notes
Bulk volume	Tetrahedral (2nd order)	0.5 - 1.0 mm	Standard mesh
Central bore	Refined hex/wedge	0.2 mm	Stress concentration
Helix threads	Swept hex (if possible)	0.3 mm	Capture curvature
Node pockets	Local refinement	0.2 mm	Sharp features

Total element count target: 150,000 - 500,000 elements

Mesh Quality Checks:
- Aspect ratio: <5:1 preferred
- Jacobian: >0.6
- No highly distorted elements near bore

1.6 BOUNDARY CONDITIONS

For Modal Analysis (Free-Free):
- **No constraints applied** (simulate free vibration in air)
- This identifies natural frequencies and mode shapes
- Rigid body modes (0 Hz) will appear for translations/rotations (ignore these)

For Torsional Deflection Study:
- **Fix bottom face** (all DOF constrained)
- **Apply torque** to top face: 5 N·m (example load)
- Measure angular deflection: target <0.075mm at OD

1.7 ANALYSIS SETUP

Modal Analysis:
```
Analysis Type: Modal (Eigenvalue extraction)
Frequency Range: 0 - 1000 Hz
Number of Modes: Extract first 20 modes
Solver: Block Lanczos (ANSYS) or Lanczos (Abaqus)
Damping: None (for natural frequency only)
```

Frequency Response (Optional):
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Analysis Type: Harmonic Response
Excitation: Unit impulse at bore centerline
Frequency Sweep: 100 - 1000 Hz (1 Hz steps)
Damping: Rayleigh (α=10, β=0.0001) or modal damping (ζ=0.003)
Output: Displacement FRF at 12 node locations
```

1.8 EXPECTED RESULTS

Mode Shape Predictions:
Mode #	Frequency (Hz)	Type	Description
1-6	0	Rigid body	Translations + rotations (ignore)
7	280-320	Bending	First lateral bending
8	340-380	**Torsional**	**Target mode (360 Hz)**
9	420-480	Longitudinal	Axial compression
10	520-600	Radial breathing	Radial expansion/contraction

Key Validation: Mode #8 (torsional) should be near 360 Hz

1.9 POST-PROCESSING CHECKLIST

- [ ] Export frequency table (all modes 0-1000 Hz)
- [ ] Animate mode shape for target frequency
- [ ] Plot displacement magnitude at node pocket locations
- [ ] Calculate effective mass participation (modal effective mass)
- [ ] Generate Campbell diagram (if speed-dependent)
- [ ] Export FRF curve (frequency vs. displacement amplitude)

1.10 VALIDATION AGAINST PHYSICAL TEST

Once physical unit is manufactured:

Test Protocol:
1. Mount TITAN A16 on vibration isolation table (free-free boundary)
2. Apply impulse excitation (modal hammer or piezo shaker)
3. Measure response via accelerometer or laser vibrometer
4. Record frequency response function (FRF)
5. Compare measured peaks to FEA predictions
6. Iterate FEA material properties if >10% error

PART 2: QUALITY CONTROL (QC) PROTOCOL

2.1 INSPECTION SEQUENCE

QC SHALL BE PERFORMED IN THE FOLLOWING ORDER:

```
RECEIVING INSPECTION
  ↓
DIMENSIONAL INSPECTION (CMM)
  ↓
SURFACE FINISH VERIFICATION
  ↓
MATERIAL VERIFICATION
  ↓
RESONANCE TEST (IF EQUIPMENT AVAILABLE)
  ↓
FINAL ACCEPTANCE / REJECTION
```

2.2 DIMENSIONAL INSPECTION (CMM)

Equipment Required:
- Coordinate Measuring Machine (CMM)
- Minimum accuracy: ±0.002mm
- Temperature-controlled environment (20°C ±2°C)

Measurement Points:

A. Outer Diameter (OD) - 8 Locations

Point	Angle (°)	Z-Height (mm)	Target (mm)	Tolerance
OD-1	0	5.0	Ø30.00	+0.03/−0.00
OD-2	45	5.0	Ø30.00	+0.03/−0.00
OD-3	90	5.0	Ø30.00	+0.03/−0.00
OD-4	135	5.0	Ø30.00	+0.03/−0.00
OD-5	180	5.0	Ø30.00	+0.03/−0.00
OD-6	225	5.0	Ø30.00	+0.03/−0.00
OD-7	270	5.0	Ø30.00	+0.03/−0.00
OD-8	315	5.0	Ø30.00	+0.03/−0.00

Acceptance Criteria: All measurements within tolerance

B. Central Bore - 12 Locations (Top, Mid, Bottom × 4 angles)

Point	Angle (°)	Z-Depth (mm)	Target (mm)	Tolerance
B-1	0	1.0	Ø3.03	±0.03
B-2	90	1.0	Ø3.03	±0.03
B-3	180	1.0	Ø3.03	±0.03
B-4	270	1.0	Ø3.03	±0.03
B-5	0	5.0	Ø3.03	±0.03
B-6	90	5.0	Ø3.03	±0.03
B-7	180	5.0	Ø3.03	±0.03
B-8	270	5.0	Ø3.03	±0.03
B-9	0	9.0	Ø3.03	±0.03
B-10	90	9.0	Ø3.03	±0.03
B-11	180	9.0	Ø3.03	±0.03
B-12	270	9.0	Ø3.03	±0.03

Concentricity Check:
- Establish bore centerline from B-points
- Measure deviation from OD centerline
- **Acceptance:** ≤0.02mm runout

C. Total Height - 4 Locations

| Point | Radial Position (mm) | Target (mm) | Tolerance |

|-------|---------------------|-------------|-----------|
H-1	5.0 (near bore)	10.00	±0.05
H-2	12.5 (mid-radius)	10.00	±0.05
H-3	12.5 (mid-radius, 90°)	10.00	±0.05
H-4	14.5 (near OD)	10.00	±0.05

D. Top Lattice Node Pockets - 12 Nodes

For each node (1-12), measure:
1. **X-Y Position** (verify against table in RFQ Appendix A)
2. **Pocket Diameter** (target: Ø2.00mm ±0.03mm)
3. **Pocket Depth** (target: 1.00mm ±0.03mm)

Example for Node 1:
Feature	Target	Measured	Accept?
X-coord	+5.000 mm	[FILL]	☐
Y-coord	0.000 mm	[FILL]	☐
Diameter	Ø2.00 mm	[FILL]	☐
Depth	1.00 mm	[FILL]	☐

Repeat for all 12 nodes.

2.3 SURFACE FINISH VERIFICATION

Equipment:
- Contact or optical profilometer
- Measurement length: 4.0mm (cutoff: 0.8mm per ISO 4287)

Measurement Locations:
Surface	Location	Target Ra (μm)	Accept Range
Top face	Center (3 points)	0.8 - 1.6	0.6 - 2.0
Bottom face	Center (3 points)	0.8 - 1.6	0.6 - 2.0
Bore inner	Mid-depth (2 points)	≤1.0	≤1.5
OD surface	4 locations	0.8 - 1.6	0.6 - 2.0

Acceptance: All measurements within range, no single outlier >2.5 μm

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2.4 MATERIAL VERIFICATION

A. Composition Check (If In-House Capability)
- **Method:** EDS (Energy Dispersive Spectroscopy) or XRF
- **Locations:** 3 random surface points
- **Target:** Ta, Ti, Au, C elements detected
- **Acceptance:** Presence confirmed (quantitative analysis not required)

B. Grain Size Analysis (Required)
- **Method:** SEM with EBSD (preferred) or metallographic etch + optical microscopy
- **Sample:** Sacrificial witness coupon from same billet OR edge section
- **Target:** ≤15 μm average grain size
- **Acceptance:** Mean grain size ≤20 μm (±5 μm tolerance from target)

If grain size >20 μm: Flag for engineering review (may still be acceptable)

2.5 RESONANCE TEST (OPTIONAL BUT RECOMMENDED)

Equipment Required:
- Modal hammer (PCB 086C03 or equivalent)
- Accelerometer (PCB 352C33 or similar, <1 gram mass)
- Data acquisition system (NI DAQ or SigLab)
- FFT analysis software

Test Setup:
1. **Mount:** Suspend TITAN A16 on soft foam or elastic bands (simulate free-free)
2. **Sensor placement:** Attach accelerometer to top face (wax mount)
3. **Excitation:** Strike bore centerline with modal hammer (single impulse)
4. **Acquire:** Record 2 seconds at 5 kHz sampling rate

Analysis:
1. Compute FFT of acceleration signal
2. Identify peak frequencies in range 100-1000 Hz
3. Measure Q-factor: $Q = f_0 / \Delta f$ (where Δf is half-power bandwidth)

Target Validation:
- **Primary peak:** 360 Hz ±10 Hz (350-370 Hz acceptable)
- **Q-factor:** >100 (low damping, sharp resonance)
- **Secondary peaks:** Document all (compare to FEA)

Acceptance:

- Peak exists in range 350-370 Hz: **PASS**
- Peak outside range but Q>100: **CONDITIONAL PASS** (document for analysis)
- No clear peak or Q<50: **REVIEW REQUIRED**

2.6 VISUAL INSPECTION

Magnification:
- 10× for general inspection
- 50-100× for critical features (bore entry, node pockets)

Reject Criteria:
- Visible cracks (any length)
- Delamination at layer interfaces
- Pitting or porosity >0.5mm diameter
- Scoring/gouging >0.1mm deep
- Discoloration indicating oxidation or contamination

Acceptance: No defects per reject criteria

2.7 QC TRAVELER TEMPLATE

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TITAN A16 - QUALITY CONTROL TRAVELER
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PART SERIAL NUMBER: TA16-[UNIT NUMBER]-[DATE]
INSPECTOR: _________________________ DATE: _______

┌─────────────────────────────────────────────────────────────┐
│ STEP 1: DIMENSIONAL INSPECTION (CMM)          │
├─────────────────────────────────────────────────────────────┤
│ OD (8 points):       ☐ PASS  ☐ FAIL  ☐ N/A   │
│ Bore Diameter:       ☐ PASS  ☐ FAIL  ☐ N/A   │
│ Bore Concentricity:  ☐ PASS  ☐ FAIL  ☐ N/A   │
│ Total Height:        ☐ PASS  ☐ FAIL  ☐ N/A   │
│ Node Pockets (12):   ☐ PASS  ☐ FAIL  ☐ N/A   │
│                          │
│ CMM Report Attached:  ☐ YES   ☐ NO          │
│ Notes: _________________________________________ │
```

STEP 2: SURFACE FINISH

Top Face Ra: _______ µm ☐ PASS ☐ FAIL
Bottom Face Ra: _______ µm ☐ PASS ☐ FAIL
Bore Ra: _______ µm ☐ PASS ☐ FAIL
OD Ra: _______ µm ☐ PASS ☐ FAIL

Profilometer Data: ☐ ATTACHED

STEP 3: MATERIAL VERIFICATION

Grain Size: _______ µm ☐ PASS ☐ FAIL
Composition Check: ☐ PASS ☐ FAIL ☐ N/A

SEM/Optical Images: ☐ ATTACHED
Mill Cert: ☐ ATTACHED

STEP 4: RESONANCE TEST (OPTIONAL)

Peak Frequency: _______ Hz ☐ PASS ☐ FAIL
Q-Factor: _______ ☐ PASS ☐ FAIL

FRF Plot Attached: ☐ YES ☐ NO ☐ N/A

STEP 5: VISUAL INSPECTION

Cracks: ☐ NONE ☐ FOUND
Delamination: ☐ NONE ☐ FOUND
Surface Defects: ☐ NONE ☐ FOUND

Photos (6 views): ☐ ATTACHED

FINAL DISPOSITION

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☐ ACCEPT - All criteria met
☐ CONDITIONAL ACCEPT - See notes
☐ REJECT - Nonconformance documented

Inspector Signature: _______________ Date: ______

Engineering Review (if required): __________________

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### 2.8 NONCONFORMANCE HANDLING

**If any inspection fails:**
1. Document specific measurements and deviation
2. Photograph defect (if applicable)
3. Contact engineering (Matthew Campbell) for disposition
4. **Do not proceed to next QC step** until resolution

**Disposition options:**
- **Accept as-is:** If deviation is minor and doesn't affect function
- **Rework:** If feature can be corrected (e.g., re-machine bore)
- **Reject:** If critical dimension is out of tolerance

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### 2.9 DATA RETENTION

**All QC records shall be retained for minimum 5 years:**
- CMM inspection reports (raw data + summary)
- Surface finish data sheets
- Material certifications
- SEM/optical images
- Resonance test FRF plots
- QC traveler (signed)
- Photographic documentation

**Format:** Digital archive (PDF) + physical copies for serial numbers

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APPENDIX: QUICK REFERENCE CHECKLIST

Before Shipping TITAN A16:

- [ ] CMM inspection complete (all dimensions)
- [ ] Surface finish verified (Ra in spec)
- [ ] Grain size analyzed (<20 μm)
- [ ] Resonance peak documented (if equipment available)
- [ ] Visual inspection (no defects)
- [ ] QC traveler signed and dated
- [ ] All documentation packaged with unit
- [ ] Photos taken (6 standard views)
- [ ] Packaging protects from damage (foam inserts)
- [ ] Tracking number recorded

Ship when all boxes checked.

END OF FEA SETUP GUIDE & QC PROTOCOL